Final Year Project Roadmap

AI-Driven Burn Assessment & Recovery Monitoring System

Start Date: 23/06/2026 **Duration:** 14 Weeks **Document Status:** Official Scope & Plan

PROJECT EXECUTION TIMELINE

#	PHASE	TIMELINE	KEY DELIVERABLES & ACTION ITEMS
1	**Literature Review**	Week 1	• Study existing burn assessment systems and state-of-the-art computer vision models. • Chalk out current limitations, clinical gaps, and datasets used in existing literature. • Shortlist potential deep learning architecture candidates.
2	**Dataset Collection**	Week 1 – 2	• Explore and check publicly available medical/burn datasets. • Coordinate with Sir Hanif and conduct external outreach for medical data acquisition.
3	**Technology Selection**	Week 2	Define and finalize core tech stack across: Deep Learning · Computer Vision · Segmentation · Classification · VLM · Backend · Frontend · Database · Deployment
4	**Data Preparation**	Week 3	• Clean, annotate, and format collected burn image data. • Construct an automated data pipeline to streamline preprocessing and augmentations.
5	**Burn Classification**	Week 4	• Develop and train a CNN classifier to categorize the 3 primary burn degree types.
6	**Burn Segmentation**	Week 5	• Train a segmentation model to accurately locate and isolate burn regions. • Implement burn mask generation and precise burn surface area calculation.
7	**Severity Assessment**	Week 6	• Consult with medical professionals/doctors and research clinical estimation standards. • Deliver an integrated severity estimation module combining depth and area metrics.
8	**Recovery Monitoring**	Week 7 – 8	• Analyze sequential burn images over time to track healing progression.

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			• Measure area reduction, color changes, and tissue texture improvement. • Compute overall healing percentage and output healing graphs, recovery scores, and progress timelines.
9	**VLM Report Generation**	Week 9 – 10	• Analyze real clinical diagnostic reports to identify required medical metrics. • Leverage Vision-Language Models (VLM) to deliver automated, comprehensive clinical reports.
10	**Full-Stack Development**	Week 11 – 12	• Develop user-facing frontend and robust backend infrastructure to connect model APIs.
11	**Testing & Evaluation**	Week 13	• Conduct system integration testing, performance benchmark evaluations, and clinical edge-case testing.
12	**Deployment**	Week 14	• Deploy the end-to-end web/mobile application for active evaluation.

PROJECT CONSTRAINT

Primary Constraint: Dataset Availability & Quality

The single critical constraint governing the progression, accuracy, and performance of this project is the availability, diversity, and labeling quality of public and clinical burn image datasets.